

Abstract No.

Biomimetic Non-lamellar Lipid Nanoplatfoms for Targeted Glioblastoma Multiforme Therapy

Ana Amorim^{1,*}, Andreia Granja¹, and Salette Reis¹

¹LAQV, REQUIMTE, Department of Chemical Sciences, Faculty of Pharmacy, University of Porto

*Presenting author

Glioblastoma Multiforme (GBM), classified as a grade 4 tumor by the WHO, is the most common and malignant primary brain tumor in adults. Despite current therapies, recurrence happens in 75-90% of cases and survival is extremely low as patients usually have a median survival time inferior to 15 months and a 5-year survival rate below 5%.

Non-lamellar liquid crystalline lipid nanoparticles are a new generation of lipid nanoparticles that self-assemble into cubic or hexagonal structures. These complex internal architectures improve long-term stability and provide both hydrophilic and lipophilic domains that enable the incorporation of diverse therapeutic agents, their protection from degradation and sustained release.

In this study, biomimetic non-lamellar nanocarriers with tumor homing capabilities were developed for GBM therapy. The formulations exhibited suitable sizes for brain drug delivery with a good drug encapsulation efficiency. Stability was maintained for at least 3 months upon storage and for 24 h in protein-rich biological media. The nanoplatfoms were hemocompatible and a sustained drug release was achieved at pH 7.4 when compared to a free drug solution. Drug-free formulations were cytocompatible in both healthy and GBM cells, while drug-loaded formulations exhibited dose-dependent anti-cancer activity in GBM cells. The nanoparticles exhibited good permeation across an *in vitro* BBB model and higher cell uptake in GBM than in non-cancer cells, demonstrating the tumor homing ability of the developed nanoplatfoms.

Overall, the developed biomimetic nanoplatfoms show great potential for targeted GBM therapy.

Keywords: glioblastoma multiforme, non-lamellar lipid nanoparticles, brain drug delivery

Funding: This work received financial support from the PT national funds (FCT/MECI, Fundação para a Ciência e Tecnologia and Ministério da Educação, Ciência e Inovação) through the project 10.54499/2023.13674.PEX. - Glioblastoma personalized therapy based on bioinspired nonlamellar LYotropic colloid Nanoparticles with Tumor homing ability (GLYNT).



Supported by SciSerTec

Web: <http://www.scisertec.de>